1. a) Dacă o persoană a investit 1000 $ în ETH la 1 ianuarie 2019, ce este valoarea investiției la 30 Aprilie 2022?

$$V_f = V_i \times e^{N \times \mu}$$

μ → mean of daily log-returns

N → number of observations

V_i → initial investment

$$V_f = 1000 \times e^{1580 \times 0.00092} = 3669{,}37\ \$$$

b) 1. Interpretation of p-values

H_0: ETH returns follow a normal distribution

H_a: ETH returns do not follow a normal distribution

if p-value < 0.05 ⇒ we reject the normality assumption

Kolmogorov-Smirnov: compares the ECDF (empirical cumulative distribution function) with the TCDF (theoretical cumulative distribution function); A higher D-stat means a bigger diff between two of them

Cramer-von Mises: a larger W-sq means a bigger deviation from normality

Anderson-Darling: A^2_{DS} indicates a substantial deviation from normality especially in the tails (fat-tails or extreme events)

↳ the strongest, suggests heavy tails

→ Financial implications:

• Risk models relying on normality (Sharpe Ratio, VaR) may underestimate risks.

↳ other models Conditional VaR

Extreme Value Theory (EVT)

2. $$P(r_t < x) = \left(\frac{c}{|x|}\right)^{\alpha}$$

c → scale parameter

α → shape parameter

a) $\alpha = 2.84$

$c = 0.07$

$P(r_t < -0.10) = ?$ ⇒ $P(r_t < -0.10) = \left(\frac{0.07}{|-0.10|}\right)^{2.84} = 0.3631$

⇒ Probability of a drop greater than 10% is 36.31%

b) the recurrence interval in years is the inverse of the probability

$$\text{Interval (years)} = \frac{1}{P(r_t < -0.10) \times 252} =$$

≈ 0.0109 years ~ 2.7 days

A 10% daily drop in the NIKKEI 225 index is expected to occur approximately every 4 days based on the Pareto distribution.

3. a) $$R = \frac{P_t}{P_{t-1}} - 1 \rightarrow \frac{P_t}{P_{t-1}} = 1 + R$$

$$\Rightarrow r = \ln\left(\frac{P_t}{P_{t-1}}\right) = \ln(1 + R)$$

b) $$V_f = V_i \times e^{r_{total}} \qquad r_{total} = r_1 + r_2 + \dots + r_5$$

$$V_f^{Rem} = V_i \times (1 + r_1) \times (1 + r_2) \times \dots \times (1 + r_5)$$

4. The normal model dramatically underestimate tail risk. For small losses the normal distribution is inaccurate but not catastrophic.

For medium and large losses, the normal model completely fails.

Crashes happen and due to market dynamics, these are not captured in a normal model.

5. $\alpha \ln c = -14$

$\alpha = 3$

$x = -0.1$

$$\Rightarrow P(X > x) = \left(\frac{c}{x}\right)^{\alpha}$$

$$P(r_t < -0.1) = P(r_t > 0.1) = \left(\frac{c}{0.1}\right)^{\alpha}$$

$\alpha \ln c = -14 \Rightarrow \ln c = -\frac{14}{3} \Rightarrow c = e^{-\frac{14}{3}} = -4.67$

$c = e^{-4.67} \approx 0.0093$

$$P(r_t < -0.1) = \left(\frac{0.0093}{0.1}\right)^3 = (0.093)^3 = 0.000832$$

0.083%

b) $$T(r_t < -0.1) = \frac{1}{250 \cdot P(r_t < -0.1)} \approx 4.81 \text{ years}$$

c) Compute probability of two consecutive 3% drops

$P(\text{two consecutive drops}) = P(r_t < -0.03) \times P(r_{t+1} < -0.03)$

$P(A \cap B) = P(A) \cdot P(B)$